

SERVICE MANUAL



USA Instruments, Inc.

Document 2375136-7

Revision 5

**GE Signa[®] 1.5T
MARK 9000
PHASED ARRAY SHOULDER COIL**

GE Catalog Part Number: M1087SE

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Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **“damage in shipment”** written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

DOC0466844 – Global Language Requirements Specification

ПРЕДУПРЕЖДЕНИЕ (BG)	• ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК. • АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД. • НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА. • НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.
警告 (ZH-CN)	• 本维修手册仅提供英文版本。 • 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。
VÝSTRAHA (CS)	• TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE. • V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKLADU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA. • NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH. • V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHO PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.
ADVARSEL (DA)	• DENNE SERVICE MANUAL FINDES KUN PÅ ENGELSK. • HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE. • FØRSØG IKKE AT SERVICE UDSTYRET MEDMINDRE DENNE SERVICE MANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET. • MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING

(NL)

- DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.
- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.

WARNING

(EN)

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR, OR PATIENT FROM ELECTRIC SHOCK, OR FROM MECHANICAL OR OTHER HAZARDS.

HOIATUS

(ET)

- KÄESOLEV TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.
- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.
- KÄESOLEVA HOIATUSE EIRAMINE VÕIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.

VAROITUS

(FI)

- TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.
- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

ATTENTION (FR)	• CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS. • SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE. • NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS • LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.
WARNUNG (DE)	• DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE. • FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN. • VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN. • WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	• ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ. • ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ. • ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ. • ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.
FIGYELMEZTETÉS (HU)	• EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHETŐ EL. • HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE. • NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK. • EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÚTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

AÐVÖRUN (IS)	• ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU. • EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLD A VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAPJÓNUSTU. • REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓK HEFUR VERIÐ SKOÐUÐ OG SKILIN. • BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.
AVVERTENZA (IT)	• IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE. • SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE. • SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO • IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.
警告 (JA)	• このサービスマニュアルは読解力が必要です。 • サービスを担当する業者が読解力がない場合は、翻訳業者の業務を行っていただく必要があります。 • このサービスマニュアルを修理時に装着サービスを行なってください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	• 본 서비스 지침서는 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오. • 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRĪDINĀJUMS (LV)	• ŠT APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGLŪ VALODĀ. • JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGLŪ, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU. • NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS. • ŠT BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.
ĪSPĒJIMAS (LT)	• ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŪ KALBA. • JEI KLIENTO PASLAUGŲ TIEKĒJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŪ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĒ. • NEMĒGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE. • JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO, • MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.
ADVARSEL (NO)	• DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK. • HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE. • IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT. • MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.
OSTRZEŻENIE (PL)	• NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIE W JĘZYKU ANGIELSKIM. • JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA. • NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO. • NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.

ATENÇÃO (PT)	• ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS. • SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO. • NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA • O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.
ATENȚIE (RO)	• ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ. • DACĂ UN FURNIZOR DE SERVICII PENTRU CLIEȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE. • NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ȘI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE. • IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPARATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.
ОСТОРОЖНО! (RU)	• ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ. • ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД. • ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ. • НЕСОБЛЮДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.
UPOZORNENIE (SK)	• TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE. • AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA. • NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU. • ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

ATENCION (ES)	• ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES. • SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION • NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO • LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.
VARNING (SV)	• DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA. • OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER. • FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN. • OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.
DİKKAT (TR)	• BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR. • EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER. • SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ. • BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.
UPOZORENJE (SR)	• OVO SERVISNO UPUTSTVO JE DOSTUPNO SAMO NA ENGLJESKOM JEZIKU. • AKO KLIJENTOV SERVISER ZAHTEVA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN DA OBEZBEDI PREVODILAČKE USLUGE. • NE POKUŠAVAJTE DA OPRAVITE UREĐAJ AKO NISTE PROČITALI I RAZUMELI OVO SERVISNO UPUTSTVO. • ZANEMARIVANJE OVOG UPOZORENJA MOŽE DOVESTI DO POVREĐIVANJA SERVISERA, RUKOVAOCA ILI PACIJENTA USLED STRUJNOG UDARA ILI MEHANIČKIH I DRUGIH OPASNOSTI.

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the Phased Array Shoulder Coil, refer to the coil labels (as shown in *Figure 1*).



Figure 1: Coil label.

Figure 2 shows a picture of the Phased Array Shoulder Coil.



Figure 2: The Phased Array Shoulder Coil.

SHIPPING LIST – TABLE 1-1

Part Name	GE Part #	Qty
Mark 9000 Shoulder Coil GE 1.5T	U1-100256	1
Patient Pad	U1-150350	1
Phantom Holder Set	U1-150412	1
Strap Set Horizontal Field (Set of 2)	U1-150375	1
Spacer Pad	2411946	1
Operator's Guide	2422148-500	1
Service Manual	2375136-7	1

1-2 Compatibility

This coil is compatible with the Signa[®] Horizon[™] and LX Horizon[™] 1.5T.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422148.

Signa[®] LX Service Methods CD, 2160623-1.

1-4 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

Dimensions

8.5" L x 9.1" W x 12.4" H (21.7 cm x 23.2 cm x 31.6 cm)

Weight

7.0 lbs. (3.2 kg)

1-5 Theory of Operation

The physical layout of the Mark 9000 Phased Array Shoulder Coil is shown in *Figure 3*. The coil is designed for imaging the shoulder and surrounding regions. The sensitive region of the coil offers approximately 20cm field of view. The device consists of three quadrature receive coils.

The Phased Array Shoulder Coil is a receive-only coil as shown in the Schematic in the Appendix. The coil is actively decoupled from the RF transmit coil during transmit by means of a RF choking circuit. The choking circuit elements are switched on by pin diodes. The pin diode is turned on by a forward biasing DC current source supplied by the MRI system (active decoupling). When the pin diode is turned on, the impedance of the RF choking circuit becomes very high (typically above 2 kilo-ohm), compared to the other circuit elements (less than 25 kilo-ohm). These high impedance elements segregate the coil circuitry into two isolated electrical segments, preventing any current flow in the coil circuit.

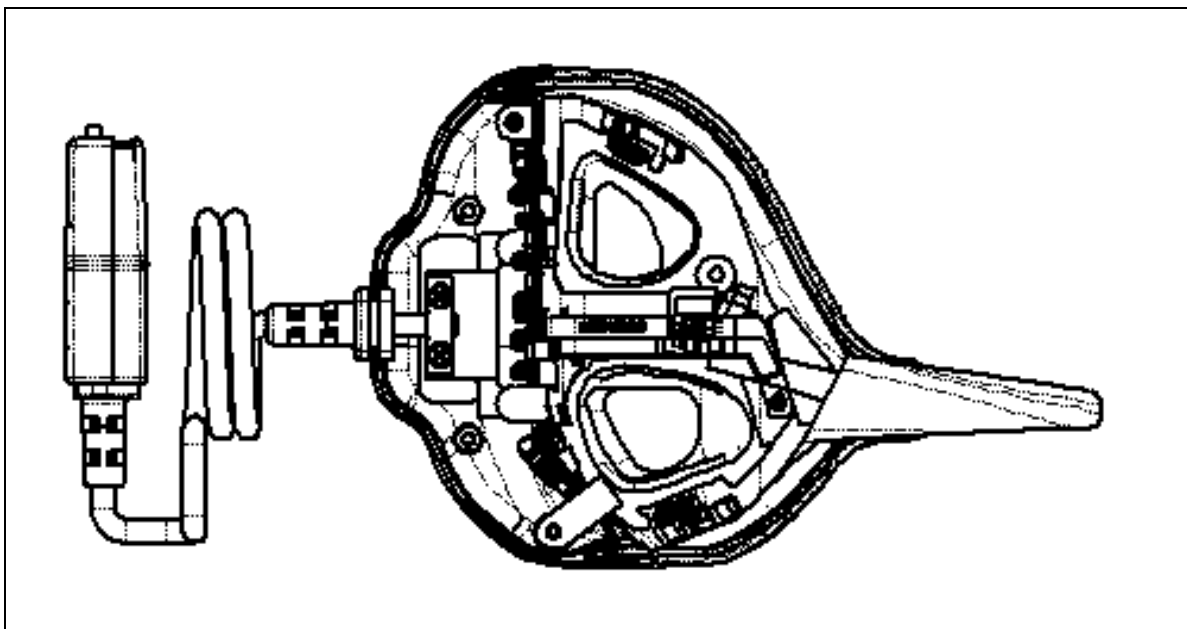


Figure 3: Coil physical layout.

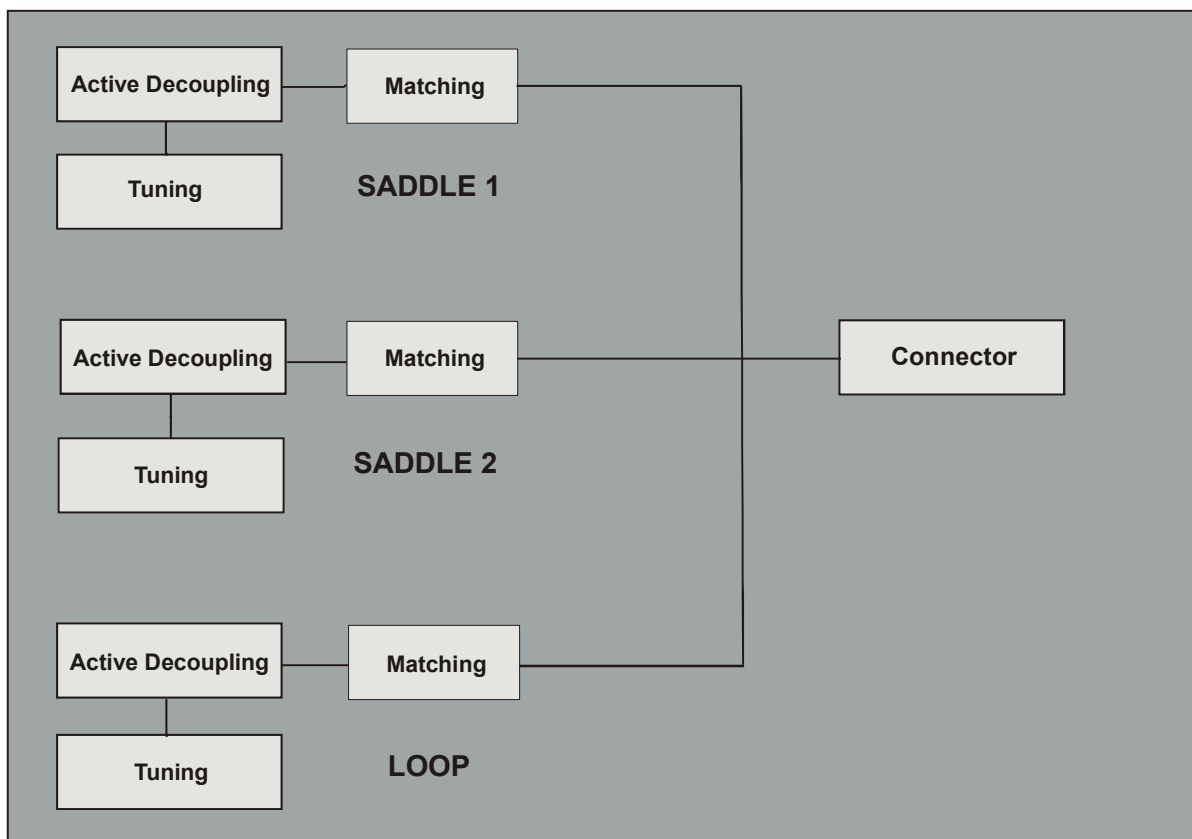


Figure 4: Block diagram of the coil.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Installing the Coil

The system will automatically recognize the coil using the Coil ID feature.

Add the coil using the Configuration File Manager. Refer to Service Methods CD; System Level Procedures; Software Utilities.

If the coil does not exist in the Coil Config File refer to the Adding New Coils to Config File Manager procedure and use the coil configuration information in Section 7-1 of this manual.

COIL CONFIGURATION NAME – TABLE 2-1-2

US SHOULDER PA

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.
2. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coils is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scope

The Functional Checks procedure in this section is intended for use on Signa software releases 9.0 and above.

3-2 Purpose

The procedure described in this section will be used to perform a quality assurance check on the Phased Array Shoulder Coil.

3-3 Overview

This procedure will walk you through collecting images and calculating SNR numbers for those images. This procedure insures the quality of the coil received at the customer site. This procedure gives detailed instructions to perform this test. The coil configuration file must be installed prior to the performance testing of this coil.

3-4 Procedure

3-4-1 Coil and Phantom Setup for the QA Scan

1. Assemble the phantom positioner as shown in *Figure 5*.

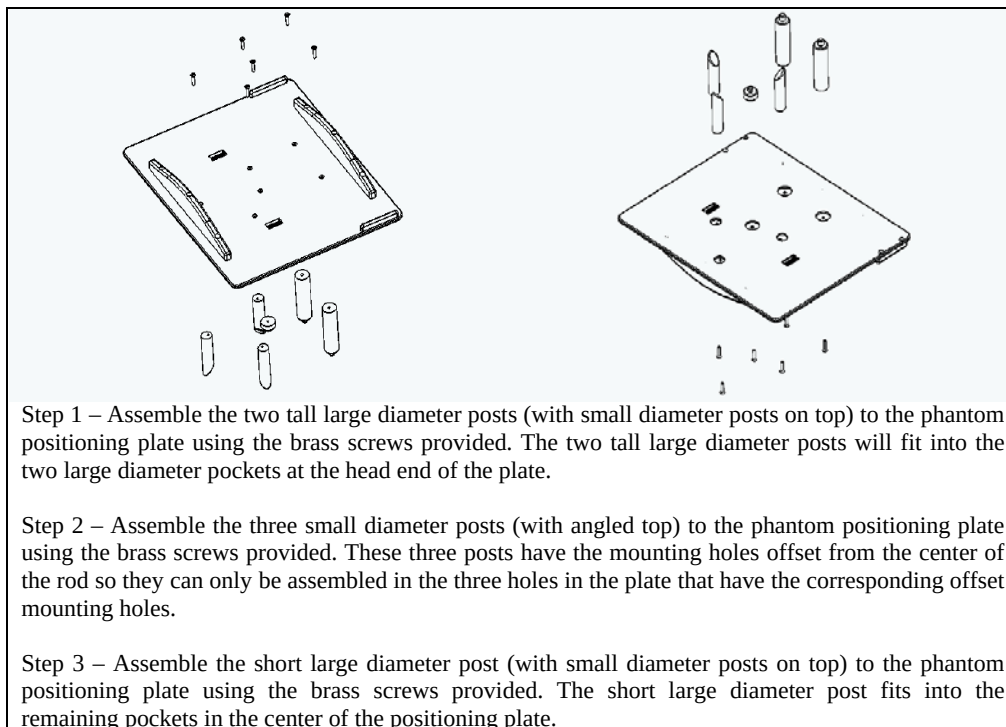


Figure 5: Assembling the phantom positioner.

2. Position the phantom positioner onto the table and place the coil and phantom onto the positioner (see *Figure 6*).



Figure 6: Positioning the phantom.

3. Landmark the coil at the crosshair.
4. At the magnet, press “**Alignment Light**” to turn on the light. Move the cradle to align the coil to the alignment lights. Press “**Landmark**” to landmark the alignment.
5. Move the coil to scan position by clicking on the “Move to Scan” button.
6. From the **Scan Desktop**, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and Patient Weight to “111” pounds. Click [**Patient Position**] to open protocols window.
7. Set the protocols using Table 3-4-1 Coronal Signal Protocol.
8. Select [**Save Series**] then [**Prepare to Scan**].
9. Prior to the scan, open the [**Display CVs**] menu under [**Research Operations**].
10. Set the “saveinter” CV to 1, saving the intermediate images. This is required to ensure all channels are operating well.
11. Select [**Auto Prescan**] then [**Scan**].
12. After the scan is complete, select [**Scan**] to run the series again.
13. Bring the table all the way out and remove the shoulder coil from the phantom positioner. Press the “Move to Scan” button to move the phantom back to isocenter.
14. Select the sequence that was just run on the shoulder coil, right click on the sequence and select [**Copy Series**]. Right click again and select [**Paste Series**].
15. Select [**View Edit**] and change the coil to Body Coil.
16. Select [**Save Series**] then [**Prepare to Scan**].
17. Select [**Auto Prescan**], after the prescan is complete, select [**Scan**].
18. After the scan is complete, select [**Scan**] to run the series again.
19. End the exam when the scan is complete. This will ensure that all CVs return to their default values.

CORONAL SIGNAL PROTOCOL – TABLE 3-4-1

Patient/Exam Information	
Patient ID	geservice
Patient Name	shoulder pa
Patient Weight	111 lbs. (50 kg)
Patient Position	
Patient Position	Supine
Patient Entry	Head First
Coil	US Shoulder Coil / Body Coil
Series Description	cor sig
Imaging Parameters	
Plane	Coronal
Mode	2D
Pulse Seq	FSE-XL
Imaging Options	none
PSD Name	<i>leave blank</i>
Protocol	<i>leave blank</i>
Scan Timing	
# of Echoes	1
TE	17
TR	500
Echo Train Length/Flip Angle	4
Bandwidth	15.63
Additional Parameters	
<i>no entries required in this area</i>	
Acquisition Timing	
Freq	256
Phase	256
NEX	1
Phase FOV	1
Preq DIR	S/I
Auto Center Freq	Peak
Autoshim	Off
Phase Correct	On
Contrast	Off
# of Reps B4 Pause	0
Scanning Range	
FOV	20
Slice Thickness	5
Spacing	1.5

	A/P	L/R Center	S/I Center
Start	A10	0	I31
End	A10		
# Slices	1	Table Delta	0.0

3-4-2 SNR Measurement

1. Subtract image four from image eight in series one, by selecting the image browser then clicking on the “add/sub” button on the right side.

NOTE: For 9.1 Lx Systems, there will be ten images taken overall. Therefore, everywhere in the SNR measurement procedure that mentions ‘Image 4’, please use Image 5. Also, everywhere it mentions ‘Image 8’, please use Image 10.

2. Select the exam, series and image eight to be the minuend. Select the “-“ minus sign in the MODE section of the window.
3. Select the exam, series and image four to be the subtrahend. Click on the equal sign in the MODE section. This will subtract image four from image eight. This image will be put into its own series, usually 100. Select the noise image in the same series as the rest of the images for display.
4. Measure with a circle, an area of 18000mm^2 plus or minus 100mm^2 on this noise image. This circle should be placed inside the phantom ball image (see *Figure 7*).

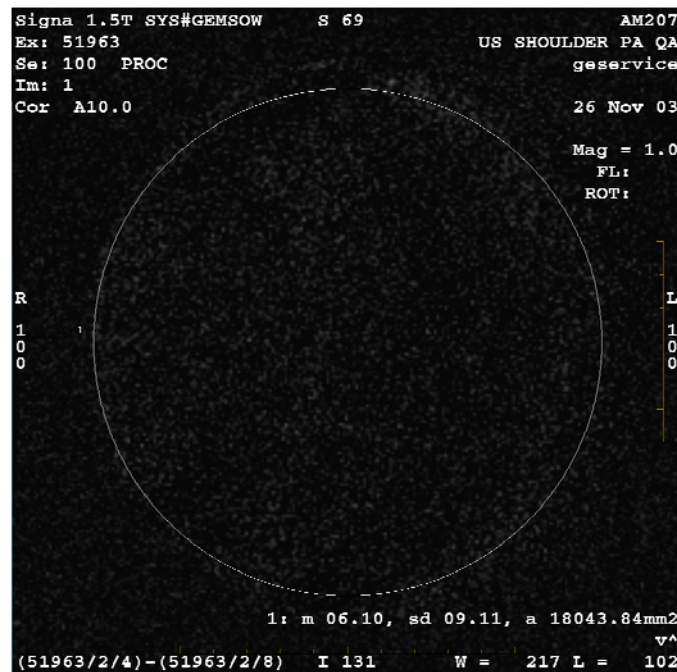


Figure 7: Subtracted noise image.

5. Record the standard deviation (sd) value in Table 3-4-2 for the three ROI's.
6. Open the Browser and select image four from the coronal signal scan.
7. Click on the “Grid” button to display the grid on image four.
8. Center the grid on signal image four, then draw three separate circles along the grid (see *Figure 8*).

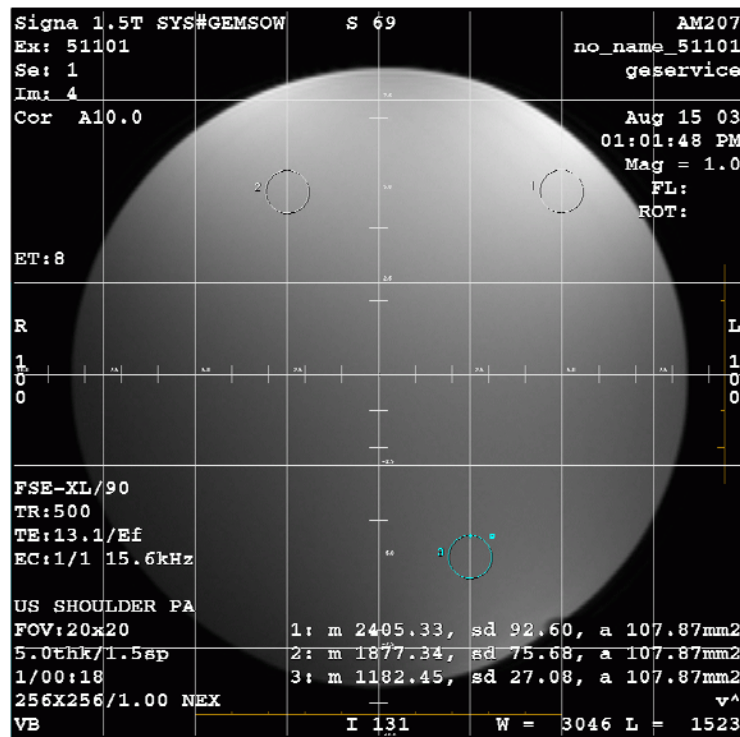


Figure 8: Signal phantom image.

9. Record the mean (m) signals in Table 3-4-2 for the three ROI's then calculate the SNR.
10. To complete the Body Coil SNR measurement, use the procedure as described above by subtracting image one from image two to create the noise scan. Measure with a circle an area of 18000mm^2 plus or minus 100mm^2 on this noise image. This circle should be placed inside the phantom ball image.
11. Record the standard deviation (sd) value in Table 3-4-2 for the three ROI's.
12. Open the Browser and select image one from the coronal signal scan body coil series.
13. Follow the same steps in the procedure above and record the mean (m) signals in Table 3-4-2 for the three ROI's then calculate the SNR.

SNR CALCULATIONS – TABLE 3-4-2

Shoulder Coil Sample	Mean 1	Mean 2	Mean 3
Noise from first image	2405.33	1877.34	1182.45
Mean Signal from the circles	9.11	9.11	9.11
SNR = Mean Signal / Noise	264.03	206.07	129.79
Body Coil Sample	Mean 1	Mean 2	Mean 3
Noise from first image	950.43	993.31	989.19
Mean Signal from the circles	44.62	44.62	44.62
SNR = Mean Signal / Noise	21.30	22.26	22.17
BODY COIL SNR	Mean 1	Mean 2	Mean 3
Noise from first image			
Mean Signal from the circles			
SNR = Mean Signal / Noise			
Tolerance for these measurements			
SHOULDER COIL SNR	Mean 1	Mean 2	Mean 3
Noise from first image			
Mean Signal from the circles			
SNR = Mean Signal / Noise			
Tolerance for these measurements			

This is not a Shoulder Coil specification. The coil must be compared against the body coil performance.

3-4-3 Performance Compared to Body Coil

The test that will be used to verify performance when compared to the Body Coil will be SNR and measured with a 15cm head phantom. SNR will be measured using the protocol described in Section 3-4-1.

3-5 Decision Criteria

The SNR of the Phased Array Shoulder Coil will be at least four times greater than the Body Coil using the same phantom, phantom holder and scan parameters.

3-6 External Cable Check

PIN Diode Check from the Connector:

Step 1 – Select the Diode Test function on the multi-meter.

Step 2 – Using *Figure 9* and Expected Readings – Table 3-3 below for the correct pin locations, connect the negative multi-meter lead to the connector pin on Row A and connect the positive multi-meter lead to the connector pin on Row B.

Step 3 – The reading on the digital multi-meter should be 0.5 ± 0.1 volts. If the reading is below 0.2 volts, either the output cable is shorted or one of the pin diodes on the feed board and the PIN diode board is defective. If the reading is above 1.0 volts, either the output cable or the DC wires are open.

EXPECTED READINGS – TABLE 3-6

Coil Element Number	Positive Lead Connection	Negative Lead Connection	Voltage Reading
Saddle 2	11B	11A	0.5 ± 0.1
Loop	9B	9A	0.5 ± 0.1
Saddle 1	7B	7A	0.5 ± 0.1
Saddle 2	11A	11A	Open
Loop	9A	9A	Open
Saddle 1	7A	7A	Open

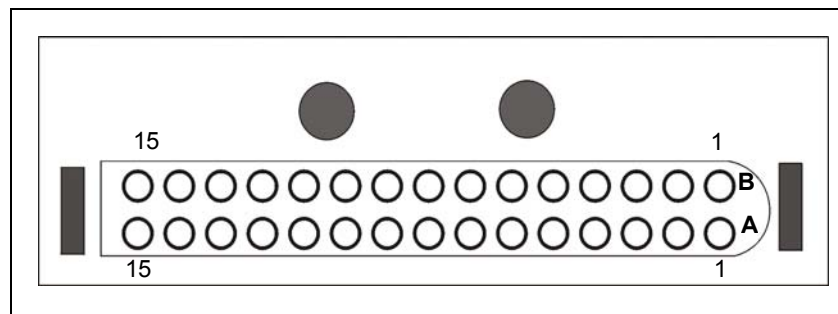


Figure 9: 30-pin Bendix connector.

3-7 PIN Diodes Check

Same as External Cable Check.

3-8 Mechanical Hardware Check

None.

3-9 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	If a short or open is observed, then replace the coil.
The output cable has a short or an open.	Disconnect the cable at the coil and check each coaxial line with a DVM (resistance). Try moving and twisting the cable as you watch the meter to find any intermittent connections. Inspect the center pins of the SMB and Bendix connections for wear and proper engagement into the mating connection.	Replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs	Correct MRI system problem.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

Coil Element Information

Element Name	Not Used	Not Used	Not Used	Saddle 2	Loop	Saddle 1	Not Used	Not Used	Not Used
Channel Number	1	2	3	4	5	6	7	8	N/A
Selected Channel	no	no	no	yes	yes	yes	no	no	N/A
Mc Bias Driver #	1	1	2	2	3	3	4	4	N/A
Active Bias Driver	no	no	no	yes	yes	yes	no	no	N/A

SECTION 4 – MAINTENANCE

4-1 Coil Care



WARNING!

Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.



CAUTION

Do not spray or pour cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 – Remove the coil covers by taking out the screws located in the covers (six on right side, three on left).

Step 2 – Remove the two screws that hold the cable clamp in position (cable is captured beneath).

Step 3 – Remove the three SMB connectors located on the tuning boards (TP1, TP2, TP3).

Step 4 – Remove the cable assembly from the coil.

Step 5 – To install the new cable assembly, start by capturing the cable beneath the cable clamp and install the two screws removed in Step 2.

Step 6 – Connect the SMB connectors to the correct PCB (Black RF-TP1, Silver RF-TP2, Brown RF-TP3).

Step 7 – Install the covers and replace the screws removed in Step 1. Make sure to capture the strain relief inside of the covers.

Step 8 – On the bad cable assembly remove the four screws that hold the interface housing together.

Step 9 – Transfer the interface housing to the new cable assembly already installed onto the coil. Replace the four screws removed in Step 8.

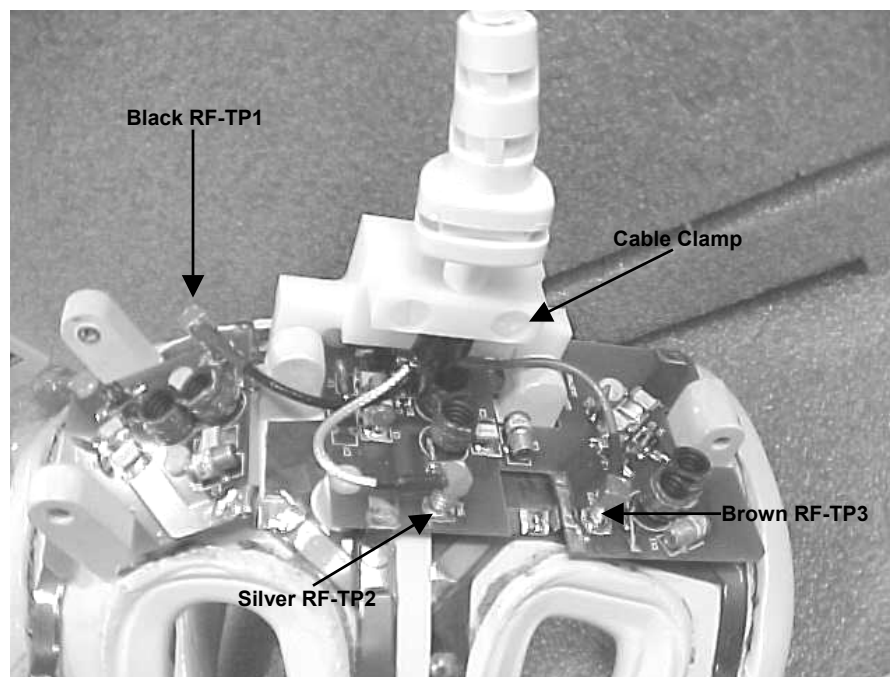


Figure 10: Cable assembly.

SECTION 6 – RENEWAL PARTS**6-1 Field Replaceable Units****FIELD REPLACEABLE UNITS – TABLE 6-1**

Part Name	GE Part #
Coil	2375136-2
Cable Assembly	2375136-8
Phantom Holder Set	2375136-3

6-2 Other Replaceable Accessories**OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2**

Part Name	GE Part #
Patient Comfort Pad	2375136-5
Positioning Straps (Set of 2)	U1-155081
Spacer Pad	2413827

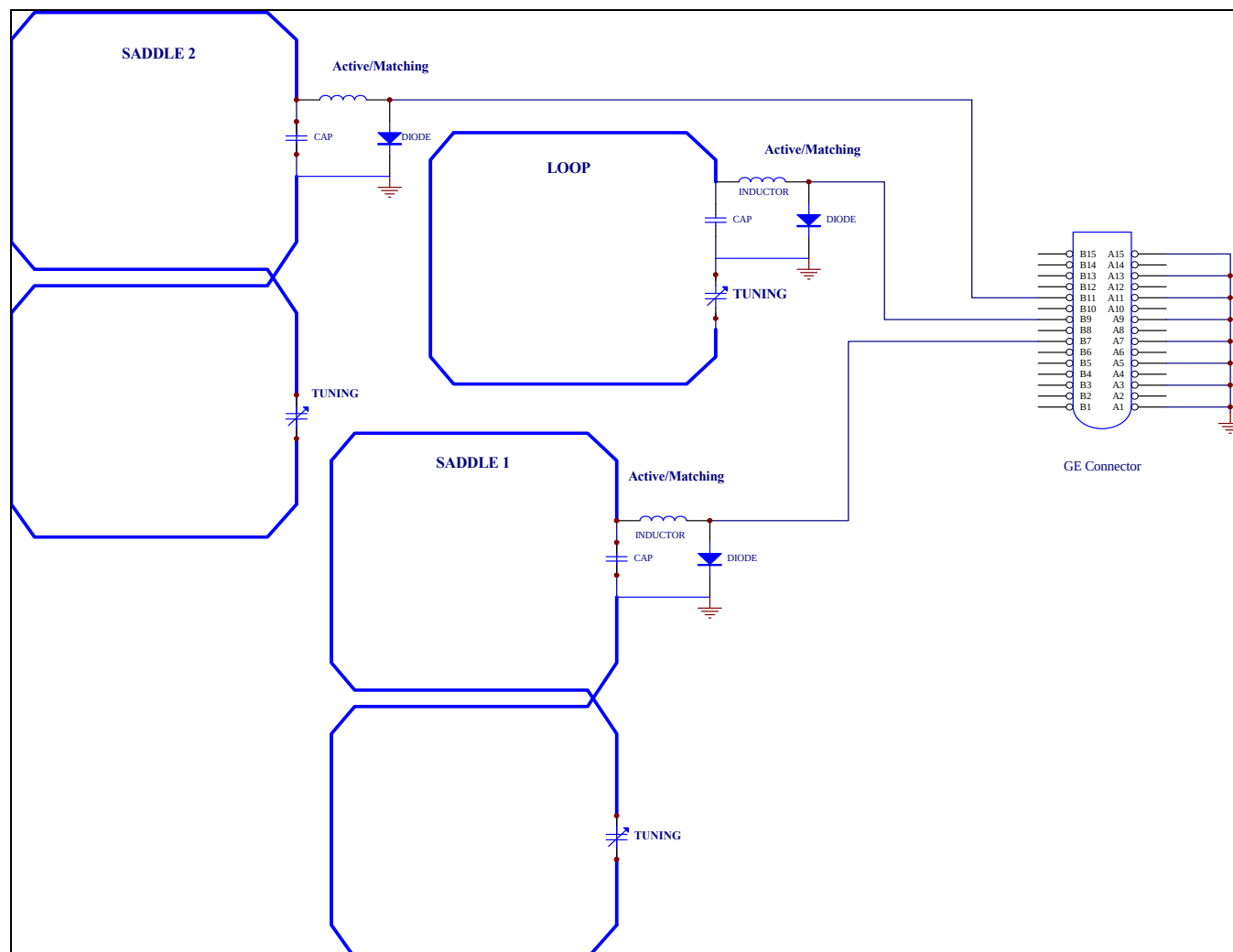
SECTION 7 – APPENDIX

7-1 Coil Configuration for Signa (These parameters are correct for release 9.1, some parameters may not be present in earlier releases.)

Parameter	Modes
Coil Name	USAI PA Shoulder
Coil Type	3
Extremity Coil	no
Cable Loss	1.05
Coil Loss	1.72
Recon Scale Factor	1
Linear vrs Quadrature	1
Multiple Receiver Coil?	yes
Number of Receivers	4
Starting Receiver ID	0
Ending Receiver ID	3
Multi-Coil Port Enable	6
Multi-Coil Port Error Enable	6
Additional Transmit Attenuation	0
Number of Fast Receivers	0
Starting Fast Receiver ID	4
Ending Fast Receiver ID	4
Start TA Value	90
End RG Value	12
Multi-Coil Recon Enable	4
Phased Array T/R Coil for Autoslim	-1
Head Default Frequency Direction	0
Axial Plane SCIC value	1.5 0.2 0.2 7.0 32 0.3 1.05
Sagittal Plane SCIC value	1.5 0.2 0.2 7.0 32 0.3 1.05
Coronal Plane SCIC value	1.5 0.2 0.2 7.0 32 0.3 1.05
Multicoil Separation Direction	0
Multicoil Switch Select	0

The configuration file for the EXCITE and EXCITE II is included in the software shipped with the coil. To install, add the coil to the site coil list.

7-2 Schematic



REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
A	04/03	Teresa DeMarco	First Issue
2	01/04	Teresa DeMarco	Revised Section 3 and General Modifications
3	07/04	J. Michael Watral	Revised Table 3-4-1(changed the slice thickness to 5mm), Table 3-6 (corrected the pin names), and Table 7-1(corrected the Port Enable and Error Enable values).
4	11/04	T. Reisker	Update coil configuration table values for release 9.x
5	06/09	Dale Seaman	Updated revision number and copyright date; Pg18 – Added note after step 1; Pg11 – Updated s/n label; Pg 11,12 – Updated Operator Manual part number; Pg3-9 – Added languages to service warning; Pg11-changed part numbers and names in Table 1-1 to match packing list; Pg 26-Added Spacer Pad to Table 6-2 and updated p/n for positioning straps.